

Package ‘jmleIRT’

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Title Joint Maximum Likelihood Estimation of the Rasch Model

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URL <https://jansteinfeld.github.io/jmleIRT/>,
<https://github.com/jansteinfeld/jmleIRT>

BugReports <https://github.com/jansteinfeld/jmleIRT/issues>

Description Implements joint maximum likelihood estimation (JMLE) for item parameters in the Rasch model (1PL). Provides methods for bias correction of item parameter estimates and weighted likelihood estimation (WLE) of person parameters.

License GPL-3

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jmlIRT-package	<i>jmlIRT: Joint Maximum Likelihood Estimation of the Rasch Model</i>
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Description

Implements joint maximum likelihood estimation (JMLE) for item parameters in the Rasch model (IPL). Provides methods for bias correction of item parameter estimates and weighted likelihood estimation (WLE) of person parameters.

Author(s)

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See Also

Useful links:

- <https://jansteinfeld.github.io/jmlIRT/>
- <https://github.com/jansteinfeld/jmlIRT>
- Report bugs at <https://github.com/jansteinfeld/jmlIRT/issues>

biasCorrection	<i>Analytical First-Order Bias Correction for Joint Maximum Likelihood Estimators in the Rasch Model</i>
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Description

This function performs an analytical bias correction for the Joint Maximum Likelihood (JML) estimators of person (θ) and item (β) parameters in the Rasch model.

Usage

```
biasCorrection(theta, beta, X, I)
```

Arguments

θ	Numeric vector of estimated person parameters $\hat{\theta}$.
β	Numeric vector of estimated item parameters $\hat{\beta}$.
X	Numeric matrix of responses (persons by items), coded as 0/1, with NA allowed for missing responses.
I	Integer scalar representing the number of items (used for bias scaling).

Details

The JML estimators are well-known to suffer from incidental parameter bias when the number of items (I) is finite, resulting in biased parameter estimates. This bias is generally of order $\mathcal{O}(1/I)$.

The bias correction implemented here follows the first-order analytic correction approach, which approximates the bias term $\mathcal{O}(B)$ for each parameter as:

$$\hat{\theta}^{corr} = \hat{\theta} - \frac{1}{I} \hat{B}_{\theta}, \quad \hat{\beta}^{corr} = \hat{\beta} - \frac{1}{I} \hat{B}_{\beta},$$

where $\mathcal{O}(B)$ is estimated based on the observed scores $\mathcal{O}(u_i)$ and the expected Fisher information $\mathcal{O}(v_i)$, calculated as

$$\hat{B}_i \approx \frac{E[v_i u_i]}{E[v_i^2]}.$$

In this implementation, scores and information are calculated based on the logistic form of the Rasch model, where $\mathcal{O}(p_{ni} = \frac{e^{\theta_n - \beta_i}}{1 + e^{\theta_n - \beta_i}})$ is the probability that person $\mathcal{O}(n)$ correctly answers item $\mathcal{O}(i)$.

Missing responses (NA) in the data matrix $\mathcal{O}(X)$ are handled by omitting those entries.

The bias correction is inspired by the incidental parameters literature (Neyman & Scott, 1948; Lancaster, 2000; Arellano & Hahn, 2006), which shows that Maximum Likelihood estimators in models with many nuisance parameters (like person parameters in Rasch) are biased when the number of observations per parameter is limited.

This function applies a computationally efficient closed-form bias correction using the first and second derivatives of the log-likelihood of the Rasch model likelihood function, evaluated at the JML estimates.

The estimator reduces bias by estimating expected score and information terms:

$$u_{ni} = X_{ni} - p_{ni}, \quad v_{ni} = p_{ni}(1 - p_{ni}),$$

for person $\mathcal{O}(n)$ and item $\mathcal{O}(i)$, and then aggregating these across items or persons.

The bias terms for person $\mathcal{O}(n)$ and item $\mathcal{O}(i)$ are estimated as:

$$\hat{B}_{\theta,n} = \frac{\sum_i v_{ni} u_{ni}}{\sum_i v_{ni}^2}, \quad \hat{B}_{\beta,i} = \frac{\sum_n v_{ni} (p_{ni} - X_{ni})}{\sum_n v_{ni}^2}.$$

This method is applicable when the number of items $\mathcal{O}(I)$ is fixed and moderate, and the number of persons $\mathcal{O}(N)$ is large.

Value

A named list containing bias-corrected parameter vectors:

theta Numeric vector of bias-corrected person parameters.

beta Numeric vector of bias-corrected item parameters.

References

- Neyman, J., & Scott, E. L. (1948). Consistent Estimates Based on Partially Consistent Observations. *Econometrica*, 16(1), 1-32. - Lancaster, T. (2000). The incidental parameter problem since 1948. *Journal of Econometrics*, 95(2), 391-413. - Arellano, M., & Hahn, J. (2006). Understanding Bias in Nonlinear Panel Models. *Review of Economic Studies*, 73(2), 557-578.

Examples

```
## Not run:
# Example data matrix with 2 persons and 4 items:
X <- matrix(c(1, 0, 1, NA, 1, 0, 1, 1), nrow = 2, byrow = TRUE)
theta <- c(0.5, -0.5)
beta <- c(-0.2, 0.1, 0.3, -0.1)
I <- ncol(X)
corrected <- biasCorrectionJMLE(theta, beta, X, I)
print(corrected$theta)
print(corrected$beta)

## End(Not run)
```

estimate_rasch_model	<i>High-level Rasch model estimation</i>
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Description

User-friendly front-end that currently dispatches to JML; kept for backward-compatibility and future method expansion.

Usage

```
estimate_rasch_model(X, method = "jml", ...)
```

Arguments

X	Numeric response matrix (0/1, NA allowed).
method	Character, currently only "jml" supported here.
...	Passed to jmle_estimation().

Value

List as from jmle_estimation().

jmle_estimation	<i>Fit Rasch model by JML with optional WLE</i>
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Description

Fits the Rasch (1PL) model using joint maximum likelihood via an efficient C++ Newton-Raphson implementation. Missing responses (NA) are allowed. Optional epsilon adjustment helps avoid biased estimates due to extreme-score degeneracy, and centering ensures identifiability. Optionally returns Warm's WLE person estimates computed with the estimated item difficulties.

Usage

```
jmle_estimation(
  X,
  max_iter = 1000,
  conv = 1e-06,
  eps = 0,
  bias_correction = FALSE,
  center = "items",
  max_update = 1.5,
  verbose = FALSE,
  estimatewle = FALSE,
  wle_adj = 1e-08
)
```

Arguments

<code>X</code>	Numeric matrix of responses (0/1) with possible NA for missing.
<code>max_iter</code>	Maximum NR iterations in C++ (default 1000).
<code>conv</code>	Convergence threshold on maximum absolute parameter change.
<code>eps</code>	Epsilon adjustment for extreme person scores (default 0).
<code>bias_correction</code>	Logical, simple post-hoc scaling of item betas.
<code>center</code>	either "items" or "persons"; enforce mean(beta)=0 or mean(theta)=0.
<code>max_update</code>	Step-size clip per update to stabilize NR (default 1.5).
<code>verbose</code>	Logical, periodic progress from C++ (every 10 iters).
<code>estimatewle</code>	Logical, compute Warm's WLE per person using estimated beta.
<code>wle_adj</code>	Small adjustment to avoid extreme raw scores in WLE (default 1e-8).

Value

A list with components: - `theta`: numeric vector of person parameters - `beta`: numeric vector of item difficulties (mean 0 if `center = TRUE`) - `iterations`: integer iterations used - `converged`: logical convergence indicator - `bias_correction`, `centered`: echoes of inputs - `wle_estimate`: numeric vector of WLE (if `estimatewle = TRUE`) or NA

prox_algorithm

PROX Estimation for the Rasch Model

Description

This function implements the PROX routine for Rasch models, ported from Alexander Robitzsch's R implementation to C++ for speed.

Usage

```
prox_algorithm(dat, dat.resp = NULL, freq = NULL, conv = 0.001, maxiter = 30)
```

Arguments

<code>dat</code>	Numeric matrix or data frame of binary item responses.
<code>dat.resp</code>	Numeric matrix indicating missing data (1 = observed, 0 = missing). If missing, it is generated internally.
<code>freq</code>	Numeric vector of frequencies for each response pattern. Defaults to 1 for all respondents.
<code>conv</code>	Numeric value specifying the convergence criterion for iteration. Defaults to 0.001.
<code>maxiter</code>	Integer specifying the max number of iterations. Defaults to 30.

Details

The function starts with initial estimates using standardized raw scores and applies a bias adjustment to item difficulties based on Robitzsch's formula:

$$d_i = \mu_i - \sqrt{1 + \frac{\sigma_i^2}{2.9}} \times \text{logit}(\text{item proportion})$$

to correct bias in finite samples. Then, person and item parameters are updated iteratively using logistic transformations until convergence or the maximum number of iterations is reached.

This implementation is primarily for quick and robust approximate Rasch model estimation, useful as a starting point for joint maximum likelihood estimation.

Value

A list with elements:

b Estimated item difficulties (logits).

theta Estimated person abilities (logits).

iter Number of iterations performed.

sigma.i Estimated standard deviations of abilities for persons responding to items.

sigma.n Estimated standard deviations of difficulties for items responded to by persons.

References

Robitzsch, A. (2020). *sirt: Supplementary Item Response Theory Models*. R package version 3.10-9. <https://CRAN.R-project.org/package=sirt>

Linacre, J.M. (1994). *Many-Facet Rasch Measurement*. Chicago: MESA Press.

Examples

```
## Not run:
data <- matrix(sample(0:1, 100, replace = TRUE), nrow = 10, ncol = 10)
result <- prox_algorithm(data)
print(result$b)

## End(Not run)
```

summary.jmleIRT	<i>Summary of JML estimation</i>
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Description

Print a brief summary for a JML fit, showing means of person and item parameters. Expects an object with numeric components 'theta' and 'beta'; designed for objects of class "jmleIRT".

Usage

```
## S3 method for class 'jmleIRT'
summary(object, ...)
```

Arguments

object	An object of class "jmleIRT" containing numeric vectors 'theta' and 'beta'. If not a strict fit object, it must still provide 'theta' and 'beta' accessible as 'object\$theta' and 'object\$beta'.
...	Further arguments passed to or used by methods (currently unused).
digits	Number of digits to display (default is 4).

Value

The input object, returned invisibly.

```
# Minimal example with an adapter object X <- matrix(c( 1, 0, 1, NA, 0, 1, 1, 1, 1, 0, 0, NA, NA,
0, 1, 0, 0, 0, NA ), nrow=5, byrow=TRUE)
res <- jmle_estimation(X, max_iter=200, estimatewle=TRUE, verbose=FALSE) summary(res)
```

See Also

[jmle_estimation]

wle_estimation	<i>Compute Warm's WLE for given item difficulties</i>
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Description

Convenience wrapper to compute person WLE given a response matrix and item difficulties, using the C++ implementation. NA in X are allowed.

Usage

```
wle_estimation(
  X,
  beta,
  max_iter = 100,
  tol = 1e-10,
  lower_ext = NA_real_,
  upper_ext = NA_real_
)
```

Arguments

<code>X</code>	Numeric response matrix (0/1) with possible NA.
<code>beta</code>	Numeric vector of item difficulties, length <code>ncol(X)</code> .
<code>max_iter</code>	Max iterations for person-level NR (default 100).
<code>tol</code>	Convergence tolerance on NR step size (default 1e-10).
<code>lower_ext, upper_ext</code>	Optional finite bounds for extreme raw scores.

Value

List with `raw_score`, `wle`, `standard_error`, `conv`, `iterations`.

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